

## Chapter 3

# Material engineering in hafnia-based RRAMs: linking structural phases, stoichiometry and dielectric function

This chapter delves into the structural and dielectric properties of  $r$ -HfO<sub>1.7</sub>-based Resistive Random Access Memory (RRAM), an emerging non-volatile memory technology. By investigating the phase transitions and stoichiometric variations intentionally induced by controlled environmental exposure, this work provides insights into how these structural changes affect the dielectric properties, which critically influence device performance.

Combining simulations with experimental techniques, we reveal how oxygen vacancies and phase transformations contribute to the unique behaviour of  $r$ -HfO<sub>1.7</sub>. These findings emphasise the importance of material engineering in optimising device performance and pave the way for further advancements in RRAM technology.

### 3.1 HfO<sub>2</sub> based RRAM: importance and application

Hafnium Dioxide (HfO<sub>2</sub>), also known as hafnia, is a material of significant technological importance due to its unique properties, particularly for future microelectronic applications [1]. It possesses a high dielectric constant due to the large permittivity and wide band gap [2–5], making it a crucial material in the semiconductor industry, particularly in transistors. Its exceptional neutron absorption capabilities make it an ideal material for control rods in nuclear reactors as well [6]. Additionally, its high-temperature stability makes it an excellent candidate for thermal barrier coatings in turbine engines [7].

Beyond these applications, HfO<sub>2</sub> also plays a pivotal role in the development of next-generation memory technologies, particularly in resistive random access memory (RRAM) devices. These HfO<sub>2</sub>-based RRAM devices are increasingly used in Internet of Things applications owing to their low power consumption, scalability, non volatility, easier fabrication, fast switching speeds and long data retention time [8–11]. They perform well in high-temperature environments and are suitable for systems like vehicle-to-vehicle (V2V) communication, industrial sensors, wearable devices, and smart-home technologies [12]. Their seamless integration with CMOS technology makes them ideal for embedded memory solutions. Moreover, RRAM devices are valuable for computing tasks due to their low power consumption, low operating voltages, and their potential to meet the requirements of emerging embedded memory applications by efficiently integrating memory and processing functions. [13]. Their durability and flexibility ensure consistent performance even under mechanical stress, making them particularly suited for wearable electronics [14]. Furthermore, HfO<sub>2</sub>-based RRAM devices are being actively explored for neuromorphic computing, where they mimic synaptic functions, enabling the development of artificial neural networks and spiking neural networks [15].

### 3.2 Role of oxygen vacancies in hafnia-based RRAM

A key factor influencing the performance of hafnia-based RRAM is the presence and distribution of oxygen vacancies. Oxygen vacancies are critical for the formation and

stability of conductive filaments. The distribution and concentration of these vacancies significantly affect the resistive switching behaviour and device performance [14, 16–19]. A basic RRAM design consists of a simple metal-insulator-metal structure, with the  $\text{HfO}_2$  layer sandwiched between two metal electrodes as shown in Fig. 3.1. An asymmetric distribution of oxygen vacancies at the  $\text{HfO}_2/\text{TiO}_2$  interface enhances the performance of flexible RRAM devices, contributing to high durability and mechanical flexibility [14]. Modulating the oxygen vacancy profile through techniques, such as adjusting oxidant pulse time during atomic layer deposition, can alter the resistive switching behaviour, switching mechanisms, and current transport mechanisms in such RRAM devices [16, 20]. It has also been proven that engineering the oxygen stoichiometry in  $\text{HfO}_x$  based devices can stabilise filament formation and enable multiple switching modes, contributing to improved device performance and reliability [21]. The structural integrity and chemical composition of the layers in RRAMs are crucial as they dictate the dielectric properties, which directly influence key performance metrics such as switching speed, endurance, retention, and energy efficiency. These dielectric properties play a vital role in determining how the material stores and dissipates electrical energy, which is essential for consistent filament formation and disruption. If the dielectric material has unsuitable properties, it can lead to challenges such as leakage currents, switching variability, or even premature device failure. Moreover, the uniformity of the dielectric layers significantly impacts the filament formation process, ultimately affecting the reliability and scalability of RRAM devices.

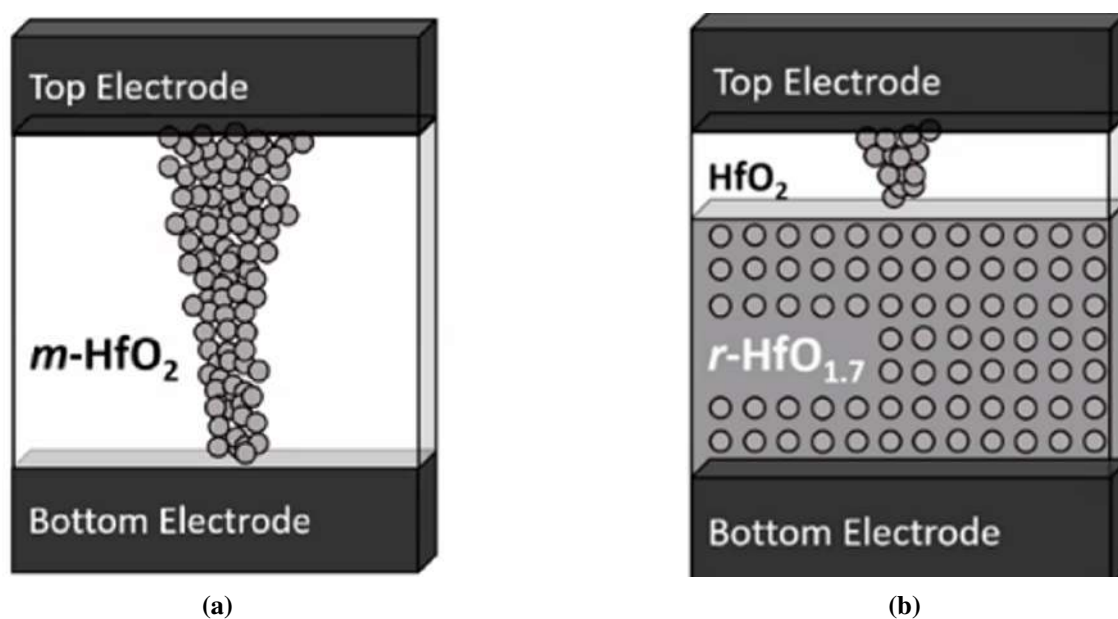
In this study, we start by understanding two  $\text{HfO}_x$  based device's fabrication techniques and experimental readings, followed by simulating in DFT the oxygen vacancies to reveal their effects on the dielectric function.

### 3.3 Fabrication of RRAM

The samples were prepared by the Advanced Electron Microscopy Group at Technische Universitat Darmstadt. Two types of  $\text{HfO}_x$  based devices were fabricated, namely monoclinic (m- $\text{HfO}_2$ ) and rhombohedral (r- $\text{HfO}_{1.7}$ ) as shown in Fig. 3.1.

For the deposition technique, all the thin films were grown using a Molecular Beam Epitaxy (MBE) system with a base pressure ranging from  $10^{-8}$  to  $10^{-9}$  mbar. Device fabrication was carried out on C-cut sapphire substrates. Hafnium layers were deposited using metallic hafnium and a controlled flux of oxygen radicals, generated at a RF plasma power of 200 W, with the substrate maintained at a temperature of 320 °C.

Monoclinic hafnia was grown at a hafnium deposition rate of 0.1 Å/s with an oxygen flux of 0.18 sccm and the rhombohedral hafnia was deposited at a hafnium rate of 0.9 Å/s, with a lower oxygen flux of 0.1 sccm, while being exposed to air. Titanium nitride (TiN) layers were deposited at a substrate temperature of 800 °C using a Ti source with an evaporation rate of 0.2 Å/s and an RF plasma power of 340 W. Substrate temperature was monitored and calibrated using a thermocouple.



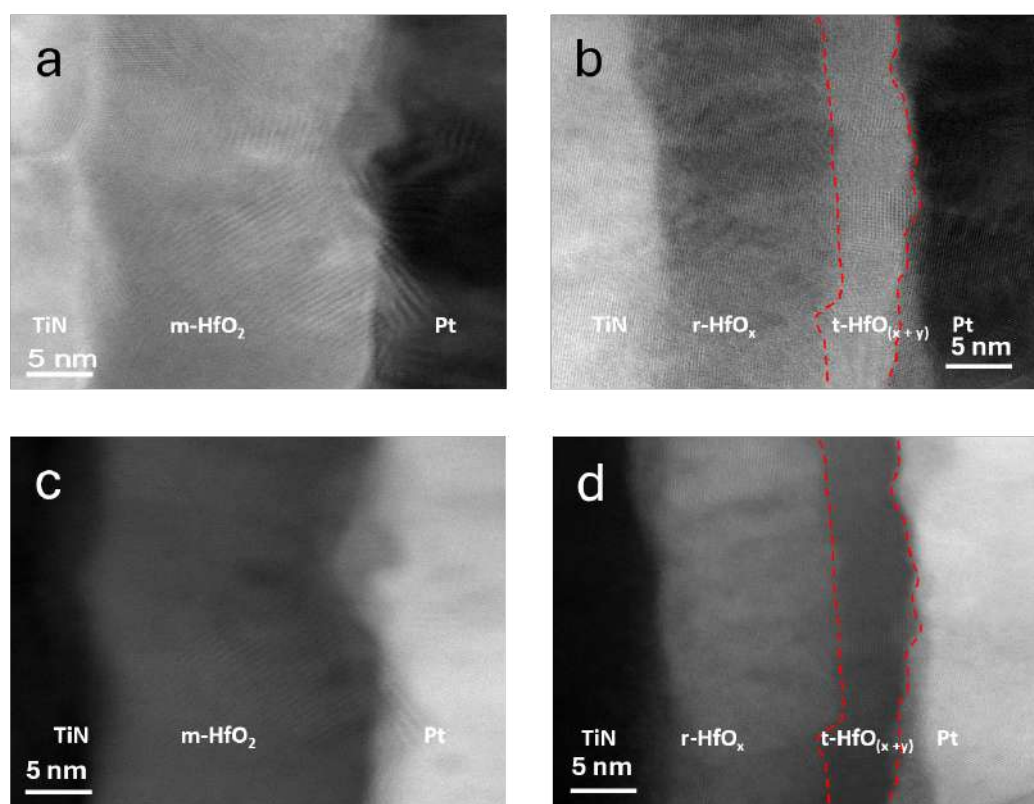
**Fig. 3.1** HfO<sub>x</sub> device layout. (a) monoclinic (m-HfO<sub>2</sub>), (b) rhombohedral (r-HfO<sub>1.7</sub>). For both devices, bottom electrode is TiN and top electrode is Pt. The extent of the potential formation of a conductive filament in the hafnia is also represented.

### 3.4 Preliminary characterisation

For each of the devices, cross-sectional TEM lamellas were prepared with a JEOL 4600F Focused Ion Beam (FIB) instrument. The lamellas were investigated with the JEOL-ARM200F electron microscope operated at 200 kV.

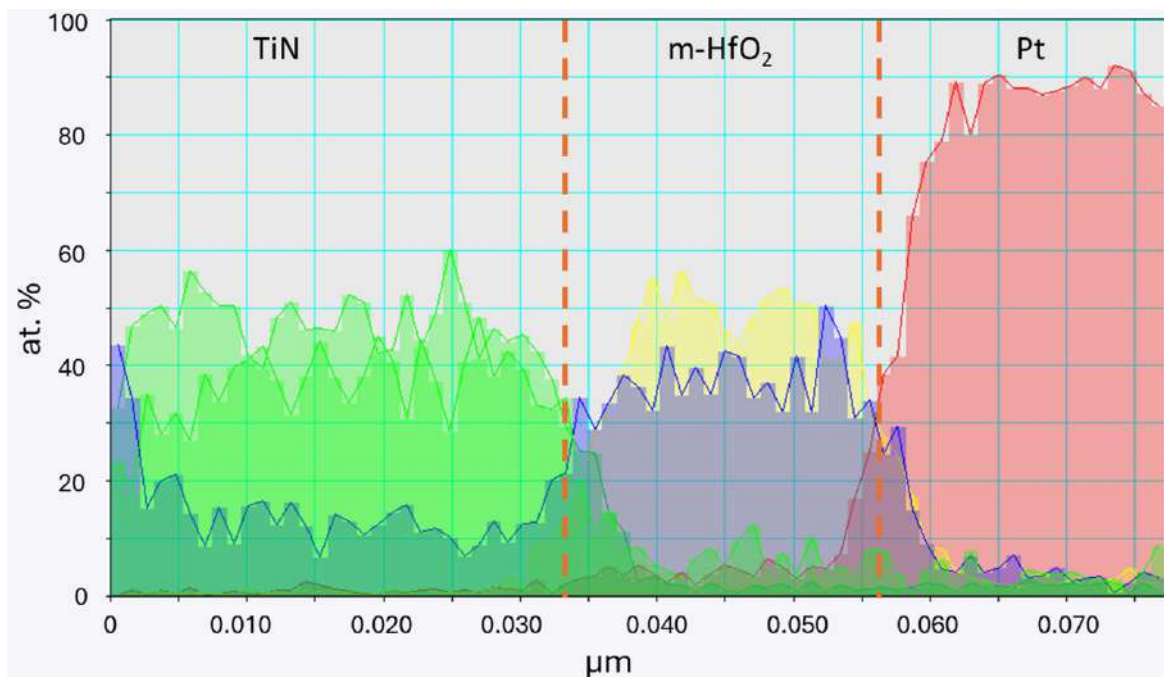
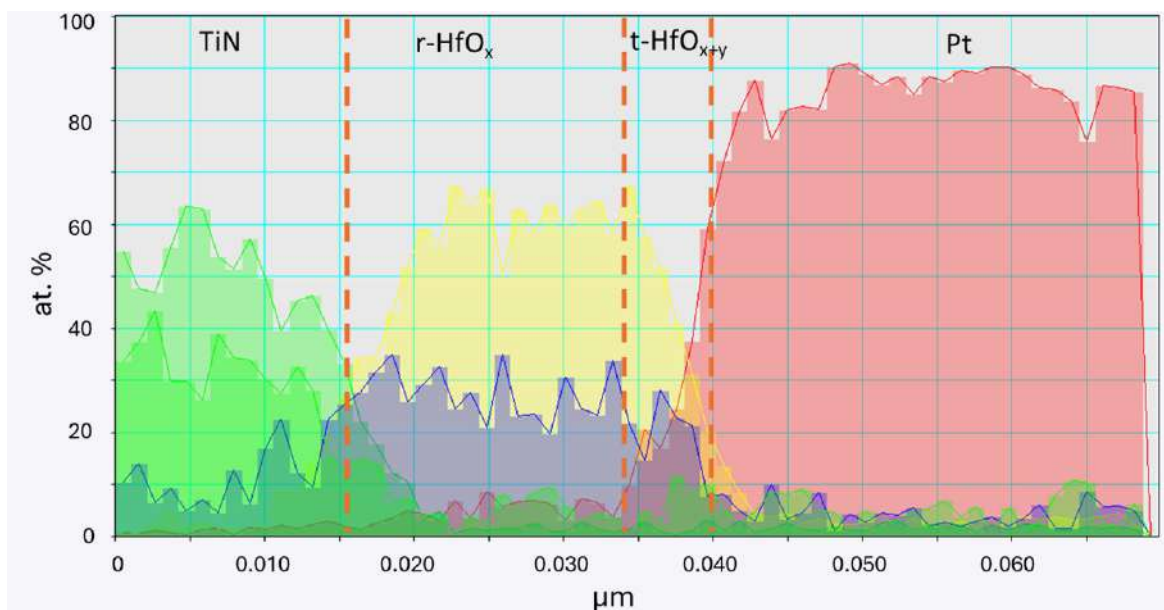
In High Angle Annular Dark Field (HAADF) images, the image intensity is directly proportional to  $Z$  raised to a power  $\alpha$ , typically ranging between 1.4 - 2 [22, 23], heavier elements, with higher atomic numbers, give rise to higher intensities. On the contrary, Annular Bright Field (ABF) images can reveal light element details that are not visible in HAADF images [24]. The HAADF and ABF images of m-HfO<sub>2</sub> and r-HfO<sub>1.7</sub> are shown in Fig. 3.2. Both devices exhibit distinct structural features, including clearly identifiable layers of titanium nitride (TiN), a hafnia layer, and platinum. However, a notable difference arises in the case of r-HfO<sub>1.7</sub>. In this material, the hafnia layer appears to be separated into two distinct regions with varying intensities, one exhibiting a higher intensity and the other a lower one. Specifically, in Fig. 3.2(b) and Fig. 3.2(d), the image of r-HfO<sub>1.7</sub> provides a clear distinction between the two sub-layers, a predecessor layer (r-HfO<sub>x</sub>) and a top layer, here referred to as t-HfO<sub>x+y</sub> within the hafnia region, showcasing the contrast variation related to differing oxygen content, since in the case of r-HfO<sub>1.7</sub>, the material was exposed to air, probably leading to partial oxygenation. In contrast, m-HfO<sub>2</sub> was prepared and maintained under vacuum conditions, which prevented this segregation and maintained a uniform oxygen distribution across the layer, as confirmed by the uniform contrast observed in its ABF image (Fig. 3.2(a)).

For further validation of the STEM results, the EDS measurements for m-HfO<sub>2</sub> and r-HfO<sub>1.7</sub> are shown in Fig. 3.3. For the m-HfO<sub>2</sub> sample, the atomic percentage composition reveals a straightforward and well-defined layered structure with three distinct regions: a TiN layer at the bottom, characterised by strong titanium and nitrogen signals; a middle layer of m-HfO<sub>2</sub>, exhibiting a consistent ratio of hafnium to oxygen, indicative of the stoichiometric monoclinic phase; and a top layer of platinum. This composition aligns precisely with the expected structure of the m-HfO<sub>2</sub> sample. The clarity of these results provides a solid foundation for comparison with the more complex r-HfO<sub>1.7</sub> sample.



**Fig. 3.2** ABF (a, b) and HAADF (c, d) of m-HfO<sub>2</sub> (a, c) and r-HfO<sub>1.7</sub> (b, d). r-HfO<sub>1.7</sub> depicts two oxide layers, one being richer in oxygen t-HfO<sub>x+y</sub> than the other r-HfO<sub>x</sub>.

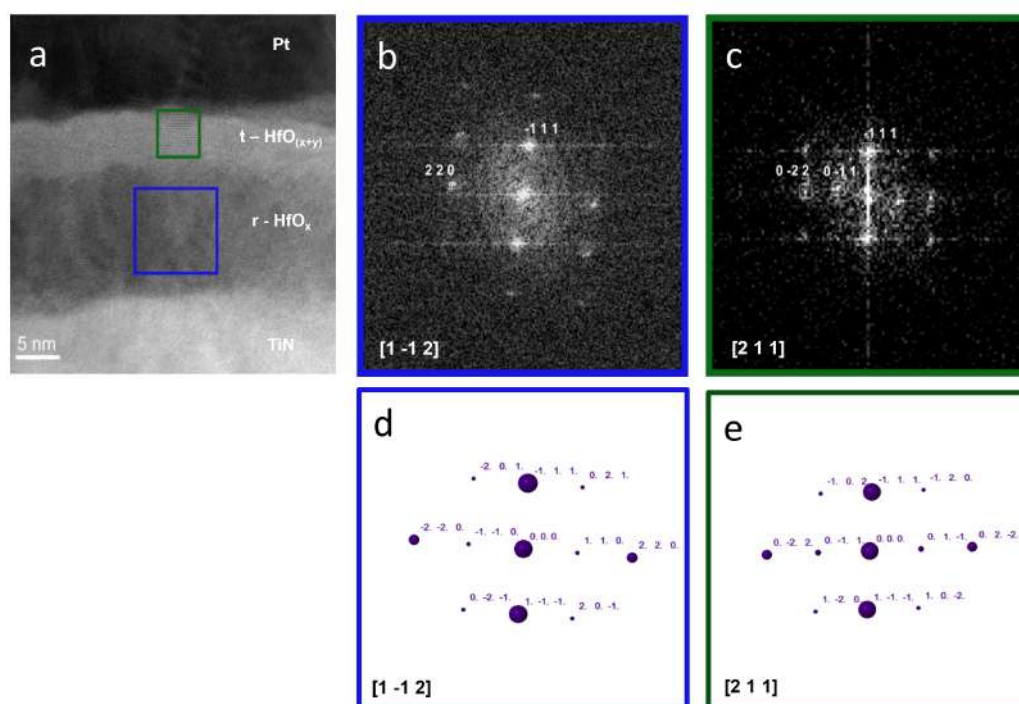
In contrast, the EDS analysis of the r-HfO<sub>1.7</sub> sample unveils a different compositional profile. While the bottom TiN layer and top Pt layer remain clearly identifiable, the hafnium oxide region exhibits a notable oxygen gradient. Near the TiN interface, a relatively stable Hf/O ratio is observed, likely corresponding to the r-HfO<sub>1.7</sub> phase. However, as the analysis progresses away from this interface, a significant change in the Hf/O ratio becomes apparent. The Hf signal decreases markedly, while the O signal shows only a slight reduction, indicating the presence of a more oxygen-rich phase, tentatively labelled as t-HfO<sub>x+y</sub>. This compositional variation suggests a possible phase transition from r-HfO<sub>1.7</sub> to a higher oxygenated layer. The EDS results, therefore, provide confirmation of the presence of a higher oxygenated layer in r-HfO<sub>1.7</sub>, which is consistent with HAADF and ABF results.

(a) m-HfO<sub>2</sub>(b) r-HfO<sub>1.7</sub>

**Fig. 3.3** Comparison of quantified EDS data for (a) m-HfO<sub>2</sub> and (b) r-HfO<sub>1.7</sub> depicting the changes in ratio of Hf to O, particularly at the interface of HfO<sub>x</sub>/Pt.

As the oxygenated layer arises only in the case of rhombohedral sample and no changes are observed in the monoclinic sample, our interest shifts to study the structural changes that are brought by the oxidation at the rhombohedral sample only.

Fast Fourier Transform (FFT) analysis was performed on the HRTEM image on the  $r\text{-HfO}_x$  and  $t\text{-HfO}_{x+y}$  regions as shown in Fig. 3.4. The observations indicate that the distances and positions of the reflections corresponding to the  $r\text{-HfO}_x$  hafnia layer (blue region) match those of the rhombohedral phase (R3m), while the  $t\text{-HfO}_{x+y}$  hafnia layer (green region) match those of the monoclinic phase (P21/c), indicating that a transformation of phase happens upon the addition of oxygen. This finding points towards a structural change from rhombohedral to monoclinic upon the oxidation of the hafnia layer and opens the door for the verification of such findings via simulation studies.

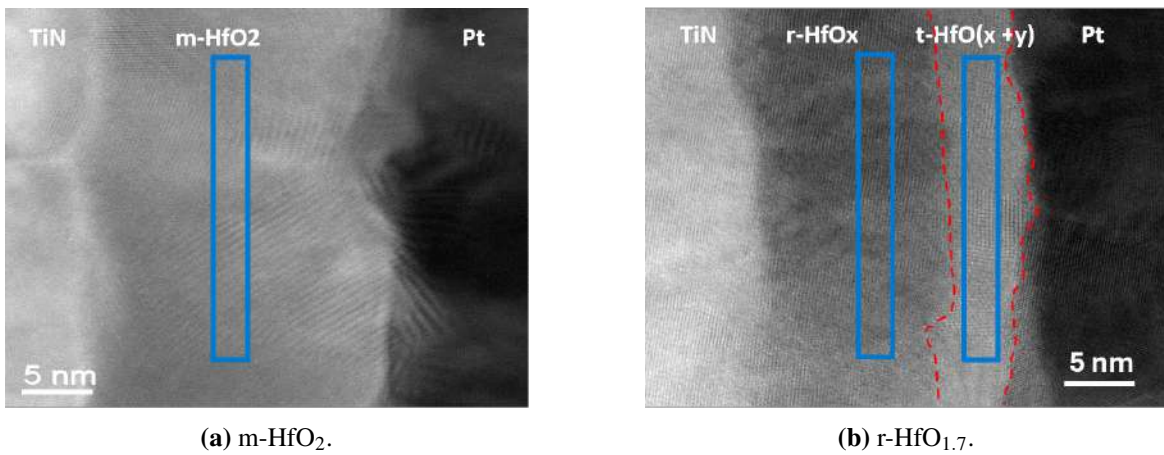


**Fig. 3.4** (a) HRTEM image for the rhombohedral hafnia sample and the FFT images for the  $r\text{-HfO}_x$  and  $t\text{-HfO}_{x+y}$  layers: (b) FFT image with zone axis  $[1 \bar{1} 2]$  of R3m symmetry, (c) FFT image with zone axis  $[2 1 1]$  corresponding to the P21/c symmetry, (d) and (e) are the theoretical diffraction patterns of (b) and (c) respectively.

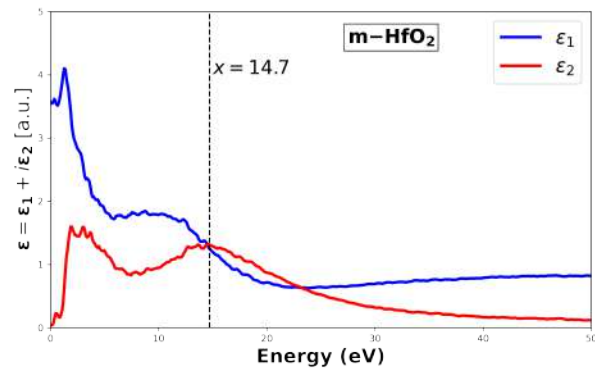
### 3.5 EELS analysis

The existence of dissimilar oxygenated layers suggests that the dielectric function of the device should change as per the structure of the concerned layer. Therefore, to study the dielectric function we chose two areas of equal dimensions on the r-HfO<sub>x</sub> and t-HfO<sub>x+y</sub> as shown in Fig. 3.5. For comparison purpose, an equal area was also chosen on the m-HfO<sub>2</sub> sample.

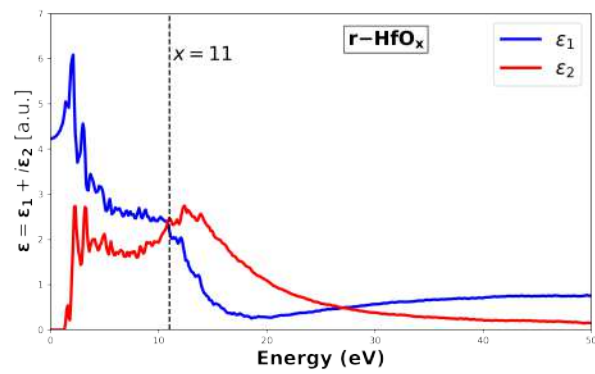
The complex dielectric function (CDF), comprising the real part ( $\epsilon_1$ ) and the imaginary part ( $\epsilon_2$ ), was computed for each area using Kramers-Kronig (KK) analysis of the EELS data, following the algorithm developed by Eljarrat [25]. Figure 3.6 illustrates the calculated CDFs, with vertical lines indicating the energy at which the real and imaginary components intersect. This intersection, a characteristic feature of the CDF, occurs at 14.7 eV for m-HfO<sub>2</sub> sample. For the r-HfO<sub>1.7</sub> sample, it is at 11 eV for r-HfO<sub>x</sub> layer, whereas for t-HfO<sub>x+y</sub>, it is shifted to 14.2 eV. This observed shift in t-HfO<sub>x+y</sub> suggests a modification of the dielectric properties due to increased oxygen concentration. Additionally, the intersection energy for t-HfO<sub>x+y</sub> closely matches that of m-HfO<sub>2</sub>. For better visualisation, separate comparisons of only the real and imaginary parts are shown in Fig. 3.7. The shape of both  $\epsilon_1$  and  $\epsilon_2$  for t-HfO<sub>x+y</sub> and m-HfO<sub>2</sub> are very similar and are different from that of r-HfO<sub>x</sub>. These results confirm that exposure to oxygen has led to a modified layer. To further evaluate and affirm our findings, simulations were performed as explained in the next section.



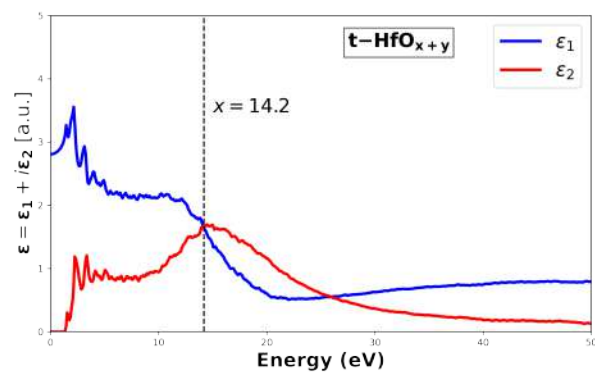
**Fig. 3.5** Selected area (in blue box) for KK analysis: (a) m-HfO<sub>2</sub>, and (b) r-HfO<sub>1.7</sub>.



(a)

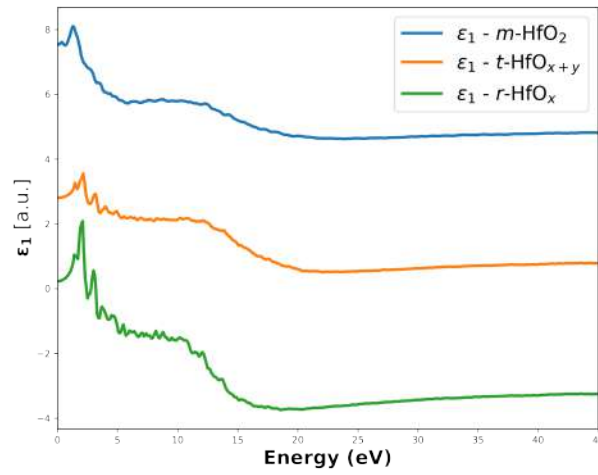


(b)

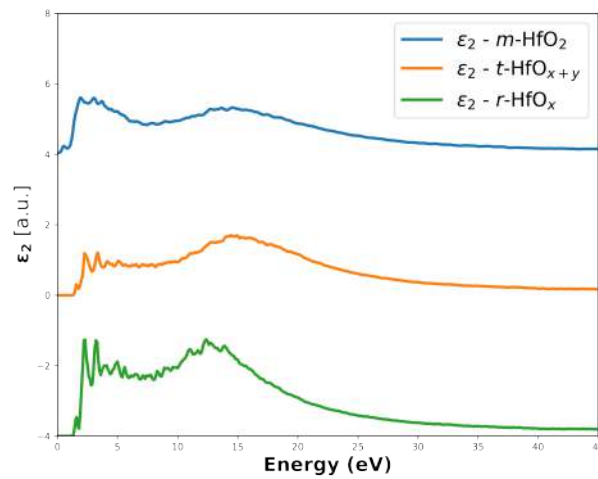


(c)

**Fig. 3.6** Dielectric function of (a) m-HfO<sub>2</sub>, (b) r-HfO<sub>x</sub>, and (c) t-HfO<sub>x+y</sub>, depicting that the characteristics of oxygenated hafnia layer are in close resemblance to that of m-HfO<sub>2</sub>.



(a)



(b)

**Fig. 3.7** Comparison of (a) real, and (b) imaginary parts of CDF for  $m\text{-HfO}_2$ ,  $r\text{-HfO}_x$  and  $t\text{-HfO}_{x+y}$ .

### 3.6 Dielectric function simulation

DFT simulations help in understanding the electronic and structural properties of hafnia and its derivatives. By providing atomistic insights, DFT enables the calculation of the CDF, offering a direct comparison with experimental results such as EELS-derived dielectric functions. In the case of  $r\text{-HfO}_x$  and  $t\text{-HfO}_{x+y}$ , DFT simulations can elucidate the impact

of varying oxygen stoichiometry on the CDF by predicting how oxidation and structural changes influence the dielectric properties.

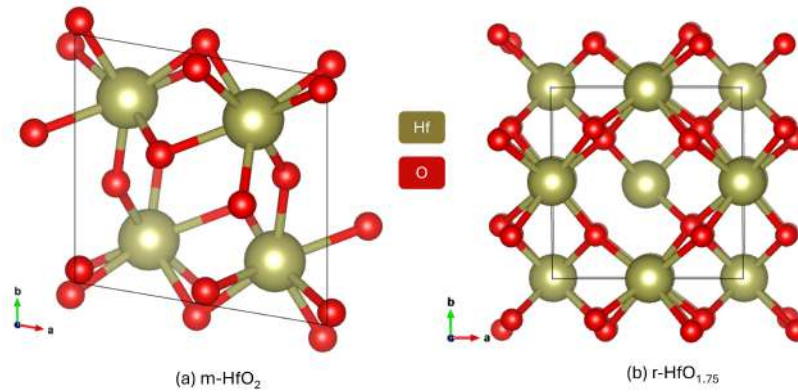
### 3.6.1 Structural modelling and oxygen vacancies

For the simulations, the two structural configurations confirmed in the samples, m-HfO<sub>2</sub> and r-HfO<sub>1.75</sub>, were considered, as illustrated in Fig. 3.8. The m-HfO<sub>2</sub> structure represents a stoichiometric form of hafnia, which is the thermodynamically stable phase at ambient conditions. In contrast, the r-HfO<sub>1.75</sub> structure was derived by systematically removing one of the eight symmetrically equivalent oxygen atoms from the cubic HfO<sub>2</sub> phase, followed by a full geometric relaxation to stabilise the resulting rhombohedral symmetry, yielding the r-HfO<sub>1.75</sub> [26].

The unit cell parameters after relaxation, for both structures are presented in Table 3.1. The m-HfO<sub>2</sub> phase exhibits an asymmetric unit cell with distinct lattice parameters and angles characteristic of a monoclinic structure. Conversely, the r-HfO<sub>1.75</sub> phase is defined by equal lattice parameters and angular distortions typical of a rhombohedral geometry.

To ensure computational feasibility while maintaining physical accuracy, the unit cells were expanded into supercells of  $2 \times 1 \times 1$  dimensions as shown in Fig. 3.9. This transformation provides a larger simulation domain, essential for accurately modelling the electronic and dielectric properties while minimising the boundary effects that might arise in smaller unit cells. DFT simulations were subsequently performed on these supercells to calculate the dielectric function for both m-HfO<sub>2</sub> and r-HfO<sub>1.75</sub>.

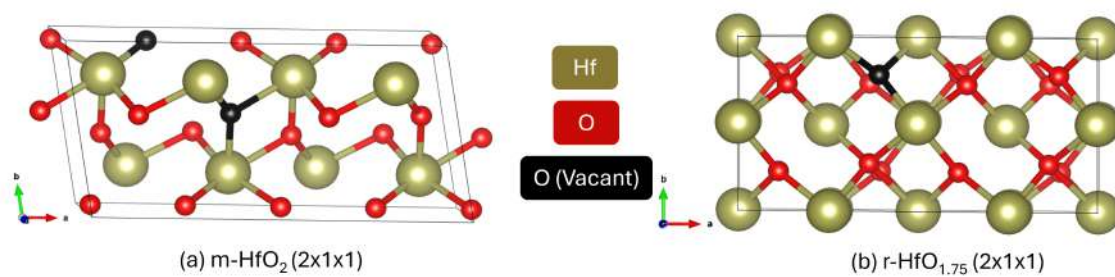
Hafnia is widely known for the presence of oxygen vacancies [27–29]. To further enhance the realism of our simulations, we introduced minor vacancies in the supercells. For both the cases at first, only one oxygen was removed from one of the non-symmetrical positions and the results were compared to that of the experimental. Further, two atoms were also removed to check the effects on the dielectric function. Here we report the best obtained results out of the multiple trials that have been performed. It has been found that the removal of one oxygen atom from r-HfO<sub>1.75</sub> and two atoms from m-HfO<sub>2</sub> as shown in Fig. 3.9 resulted in the closest match, as explained in the following sections.



**Fig. 3.8** Unit cell structure of (a) m-HfO<sub>2</sub>, and (b) r-HfO<sub>1.75</sub>.

|                       | a (Å)   | b (Å)   | c (Å)   | $\alpha$ (°) | $\beta$ (°) | $\gamma$ (°) |
|-----------------------|---------|---------|---------|--------------|-------------|--------------|
| m-HfO <sub>2</sub>    | 5.23498 | 5.07846 | 5.16442 | 90.0000      | 90.0000     | 99.6672      |
| r-HfO <sub>1.75</sub> | 5.02028 | 5.02028 | 5.02028 | 90.5356      | 90.5356     | 90.5356      |

**Table 3.1** Lattice parameters for the unit cell of (a) m-HfO<sub>2</sub> and (b) r-HfO<sub>1.75</sub>.

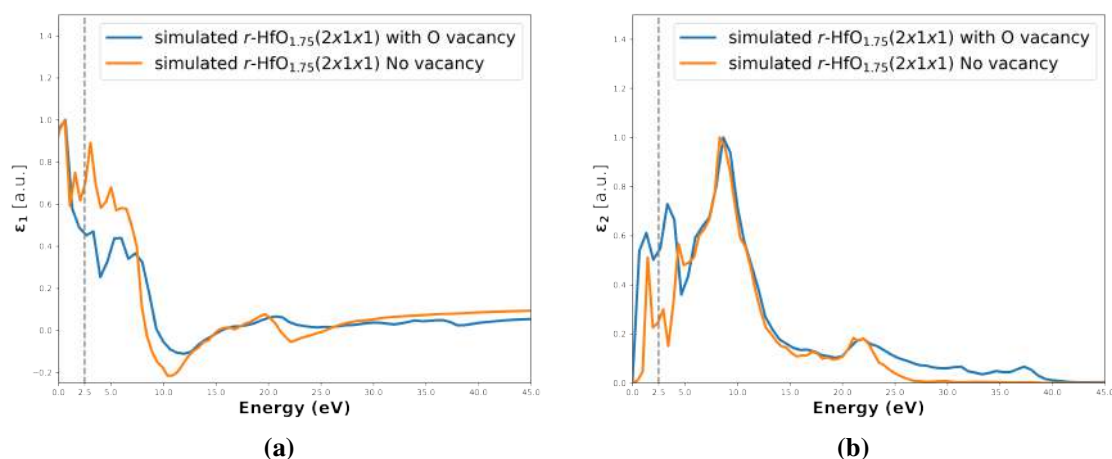


**Fig. 3.9** Supercell structure of (a) m-HfO<sub>2</sub>, and (b) r-HfO<sub>1.75</sub> with oxygen vacancies .

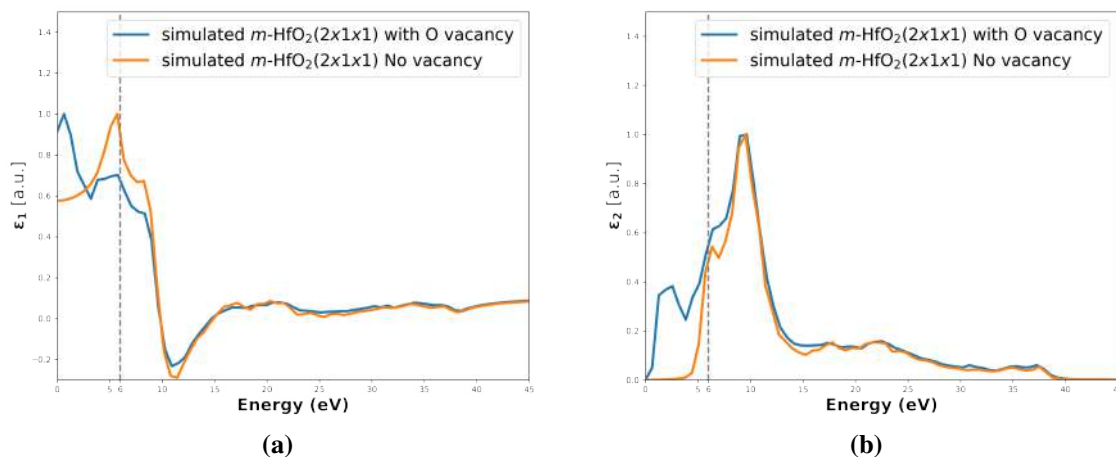
### 3.6.2 Simulation results

For each of the supercells with vacancy configurations, the structures were subjected to geometric relaxation using the Vienna Ab Initio Simulation Package (VASP) [30]. The geometry of the systems was relaxed to minimise the forces on the atoms, ensuring that the structures reached their lowest energy configuration. The Perdew-Burke-Ernzerhof [31] functional was utilised. A plane-wave energy cutoff of 500 eV was chosen. The Brillouin zone was sampled using a Monkhorst-Pack k-point mesh of  $10 \times 10 \times 10$ . The convergence criteria for the relaxation were set at  $10^{-5}$  eV. Once the relaxed structure was found, the "optics" simulation module was applied by setting the precision to accurate mode and sigma to 0.01 to obtain the dielectric properties of the structure.

The normalised simulation results of the dielectric function for supercells with and without vacancies in  $r\text{-HfO}_{1.75}$  and  $m\text{-HfO}_2$  are shown in Figs. 3.10 and 3.11, respectively. A significant difference is observed in the 0–5 eV range for all cases. In  $r\text{-HfO}_{1.75}$ , the dielectric function  $\epsilon_1$  for the vacancy-containing supercell exhibits a lower intensity around the energy of 2.5 eV compared to the vacancy-free case, while the intensity  $\epsilon_2$  is higher. Similarly, for  $m\text{-HfO}_2$ , the intensity of  $\epsilon_1$  is lower in the vacancy-containing supercell around 6 eV, while  $\epsilon_2$  shows a higher intensity around 2.5 eV compared to the vacancy-free case.



**Fig. 3.10** Dielectric function simulation ((a) real, and (b) imaginary) of  $r\text{-HfO}_{1.75}$  supercell with and without oxygen vacancy.

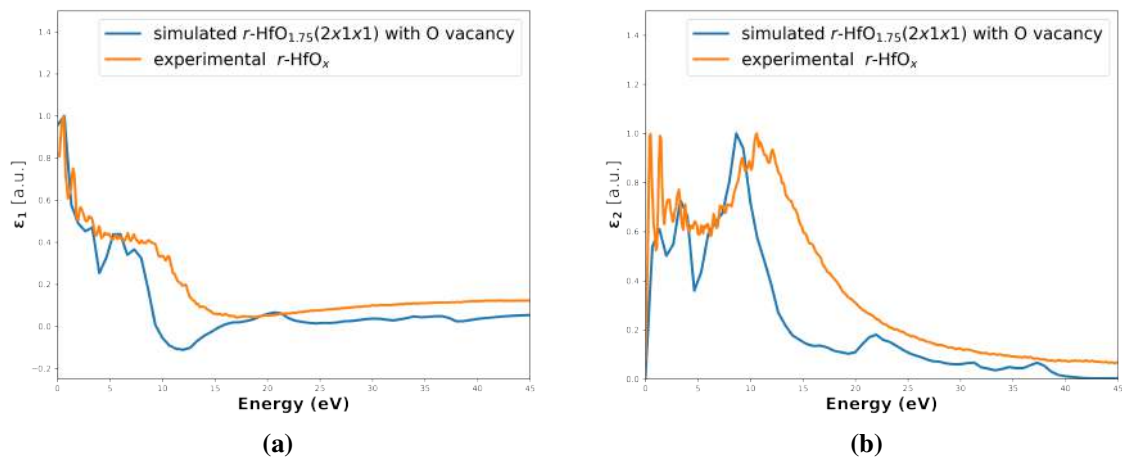


**Fig. 3.11** Dielectric function simulation ((a) real, and (b) imaginary) of m-HfO<sub>2</sub> supercell with and without oxygen vacancy.

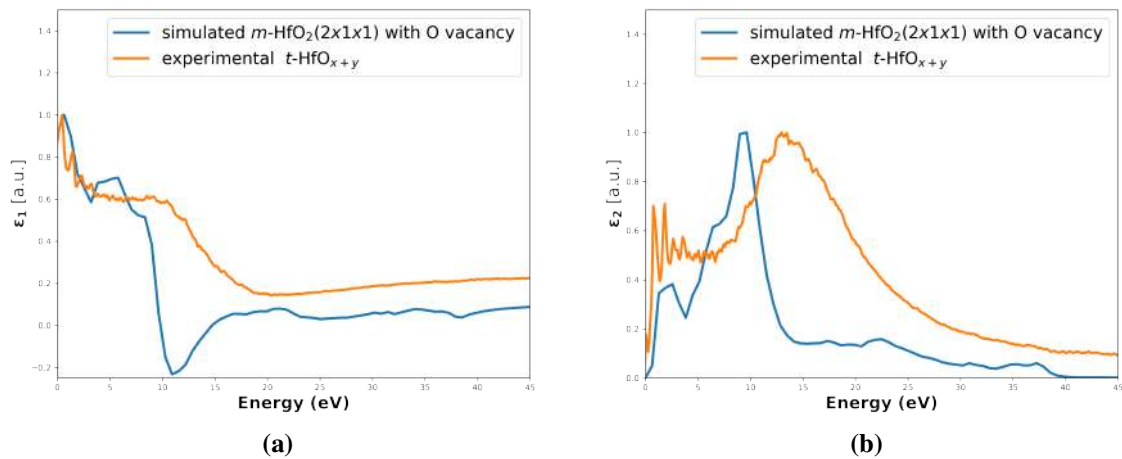
We performed a comparison of the dielectric function of r-HfO<sub>1.75</sub> with r-HfO<sub>x</sub> and m-HfO<sub>2</sub> with t-HfO<sub>x+y</sub>, as shown in Figs. 3.12 and 3.13, respectively. EELS simulated data in WIEN2K suffers a shift in the energy axis owing to various factors (relativistic calculations, broadening, onset parameters) [32–35]. Therefore, to improve the fit, the simulated plots have been shifted by 3 eV towards low energy. Overall, the simulated dielectric functions match the experimental results across all cases. The results of the simulated dielectric function of m-HfO<sub>2</sub> confirms the experimental finding that the oxygen-rich area in the rhombohedral hafnia layer undergoes a phase transition to a monoclinic phase upon oxidation. However, some minor discrepancies are noticeable, particularly around 10 eV. These differences can be attributed to several factors. First, the sample thickness was approximately 150 nm, as determined by KK calculations. This relatively large thickness could introduce effects such as multiple scattering, which may not be fully captured in the simulations, leading to slight deviations in the spectral data. Second, the exact number and ordering of oxygen vacancies in the system remain difficult to quantify with precision at this stage. Variations in the vacancy concentration and distribution significantly influence the dielectric response.

The simulated dielectric function of m-HfO<sub>2</sub> is the best match for the experimental dielectric function of the t-HfO<sub>(x+y)</sub> region, strongly suggesting that the incorporation of oxygen into the rhombohedral lattice during exposure to air induces a local transformation

favouring the formation of a monoclinic structure. This structural change is visually consistent with the ABF and HAADF images (Fig. 3.2(b) and 3.2(d), which show a clear contrast difference and distinct structural arrangement between the rhombohedral  $r\text{-HfO}_x$  region and the oxygen-rich  $t\text{-HfO}_{(x+y)}$  region, where the latter exhibits a more disordered lattice arrangement, particularly apparent at the interface between the two oxide layers in the HAADF image, suggestive of the emerging monoclinic phase. The simulation results also align with the findings of the FFT analysis in Fig. 3.4, further confirming this transition.



**Fig. 3.12** Comparison of experimental and simulated dielectric functions ((a) real, and (b) imaginary) of  $r\text{-HfO}_x$  and  $r\text{-HfO}_{1.75}$  supercells with oxygen vacancy.



**Fig. 3.13** Comparison of experimental and simulated dielectric functions ((a) real, and (b) imaginary) of  $t\text{-HfO}_{x+y}$  and  $m\text{-HfO}_2$  supercells with oxygen vacancy.

### 3.7 Discussion

This work in this chapter provides information about structural and electronic properties of hafnia-based RRAM devices and studies the effect of oxygen stoichiometry on the dielectric properties.

The evidence of a dual-layer structure in r-HfO<sub>1.7</sub> samples, compared to the uniform structure in m-HfO<sub>2</sub>, shows how the ambient environment condition affects the RRAM device structure. The rhombohedral-monoclinic phase transition with exposure to air proves that hafnia-based materials are extremely sensitive to the ambient environment.

The FFT analysis of the structural change from R3m to P21/c symmetry gives crystallographic proof that oxidation-induced phase transitions do indeed occur. This is important because it implies that device properties can be tuned locally using controlled oxidation processes.

The correspondence between dielectric response and structural phase depicts that minor variations in oxygen content affect the electronic properties. It is both an opportunity and a challenge: it facilitates fine-tuning of device parameters, yet requires delicate control over fabrication and operating conditions. The correspondence of DFT simulations with experimental EELS data supports our findings as well.

Technologically, these results indicate that oxygen-rich layer formation can be leveraged to develop engineered interfaces with designed dielectric behavior. Control of local stoichiometry of oxygen via ambient exposure or by controlled oxidation processes could be used to develop RRAM devices with enhanced switching performance, endurance, and retention characteristics.

Although this research presents useful information, certain limitations must be noted. The precise quantification of oxygen vacancy concentrations is still challenging, and the effect of other types of defects (e.g., hafnium vacancies or interstitials) on the resulting properties needs to be examined further.

## 3.8 Conclusions

In this chapter, we analyse the structural and dielectric properties of hafnia-based RRAM devices with varying oxygen content.

- Two types of  $\text{HfO}_x$  layers, monoclinic ( $\text{m-HfO}_2$ ) and rhombohedral ( $\text{r-HfO}_{1.7}$ ) were fabricated using Molecular Beam Epitaxy under different environmental conditions.
- Structural characterisation using ABF and HAADF-STEM revealed a dual-layer formation in  $\text{r-HfO}_{1.7}$ , in contrast to the uniform structure in  $\text{m-HfO}_2$ .
- EDS analysis confirmed a gradient in the Hf/O ratio in  $\text{r-HfO}_{1.7}$ , indicating an oxygen-rich top layer due to ambient air exposure, possibly linked to a phase transformation.
- FFT analysis of HRTEM images suggested that oxidation leads to a structural change from rhombohedral to monoclinic phase.
- Dielectric properties of the different layers were studied using EELS and Kramers-Kronig analysis, revealing a significant shift in the dielectric function due to increased oxygen content.
- DFT simulations with oxygen vacancies in  $\text{m-HfO}_2$  and  $\text{r-HfO}_{1.75}$  structures reproduced the dielectric trends observed experimentally.
- Simulated dielectric functions aligned well with experimental EELS results, supporting the observed phase transition and its influence on dielectric behaviour.
- The formation of an oxygen-rich monoclinic layer upon air exposure opens the potential for engineering dielectric interfaces to enhance RRAM performance.

## 3.9 References

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